Circular Economy II: PaaS & Digital Passports

Week 5: From Volume to Value Creation

Cristhian Rodríguez Revilla, PhD

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Cristhian Rodriguez
FINANCIAL ECONOMICS

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Warm-up: The CEO's Dilemma (Live Word Cloud)

Let's test what we learned in Weeks 1-4 before we dive into the Servitization pivot.

textJoin the Live Boardroom

Option A: Web
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The Dilemma

"In one word, what is the biggest financial risk of continuing to sell physical products based on maximum volume?"

Roadmap & Learning Objectives

Today's Flight Plan

1. The Problem
The Transactional Trap

2. Methodology
Servitization (CAPEX to OPEX)

3. The Tech Enabler
Digital Product Passports

The Boardroom Challenge

The Mission: Shift from selling products to selling solutions, securing long-term loyalty and regulatory compliance.

Today, you will act as the Board of Directors for either **Apple** or **Fairphone**.

Using the concepts of Servitization and Digital Passports, you must completely pivot your business model to survive the new EU "Right to Repair" regulations.

The Problem: The Incentive Paradox

The Financial Root of the Problem

We know extraction is bad for the planet. But why is the linear model actively toxic for the **balance sheet**?

The Paradox of Durability

The Flaw:
Under the linear model, if you make a product that lasts 20 years, **you destroy your own sales pipeline**.

The Perverse Incentive:
The firm is financially rewarded for creating *planned obsolescence*.

The EU Response:
The new "Ecodesign for Sustainable Products Regulation" (ESPR) heavily fine and penalize this exact behavior.

The Commodity Trap

Race to the Bottom:
If you only sell physical units, you are forced to compete entirely on price.

Information Asymmetry:
The transaction ends at the register. The customer takes the product and disappears.

The Result:
You lose all data on usage and secondary markets. **Customer Lifetime Value (CLV) drops to zero.**

The Problem: The Flaws of the Transactional Model

The Traditional Business Equation

More Revenue = Selling More Units (Volume) = **Extracting More Finite Resources**.

Margin Erosion

✖ **The Flaw:** Relying on pure volume is mathematically risky in a resource-constrained world.

✖ **The Impact:** As raw materials become scarce, production costs rise.

↓ **Result:** Profit margins collapse while environmental degradation skyrockets.

Servitization: Decoupling Growth from Resources

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" How to grow revenue while decreasing the throughput of new raw materials?
Stop selling the physical asset.
Start selling the outcome of the asset. "

The Core Strategy: Decoupling

CAPEX (Capital Expenditure)

✗ Customer pays a huge upfront cost.
✗ Goal is to *own* the product.
✗ One-time transaction. The brand loses control of the asset.

→

OPEX (Operational Expenditure)

✓ Customer pays a recurring fee.
✓ Goal is to *access/use* the product.
✓ Brand retains ownership. Profit is tied to longevity, not obsolescence.

The Financial Migration: CAPEX vs. OPEX

The Servitization Pivot

To fix the linear paradox, we change how the customer pays. We move to **PaaS (Product-as-a-Service)**.

CAPEX

Capital Expenditure (Traditional)

The Deal:
Customer pays a huge upfront cost to *own* the asset.

Cash Flow:
Sporadic, unpredictable spikes in revenue.

Firm's Motivation:
Sell it fast. Hope it breaks eventually so they buy another.

Disadvantage:
High barrier to entry. **Zero post-sale relationship.**

OPEX

Operational Expenditure (Servitization)

The Deal:
Customer pays a recurring fee to *access/use* the asset.

Cash Flow:
Predictable Annual Recurring Revenue (ARR).

Firm's Motivation:
Every repair is a cost to the firm, so they design for *extreme durability*.

Advantage:
Customer is locked into the ecosystem. **High CLV.**

The 3 Types of Product-Service Systems (Tukker, 2015)

Not all PaaS models are the same. The financial structure dictates the business risk.

1. Product-Oriented

How it works:
Sell the product, but add highly profitable services (maintenance, warranties).

Example:
AppleCare (High-margin repair ecosystem).

Risk:
Can feel like a "hidden tax" if the base product breaks easily.

2. Use-Oriented

How it works:
Provider owns the product. Client pays to access it (Leasing/Renting).

Example:
Rent the Runway (Leasing high-end fashion).

Risk:
High logistical costs (shipping, cleaning) can destroy operating margins.

3. Result-Oriented

How it works:
Client pays *only* for the final outcome. No product changes hands.

Example:
Philips "Pay per Lux" (Airports pay for light, not bulbs).

Risk:
Requires massive upfront capital (CAPEX) from the provider.

The Tech Enabler: Digital Product Passports (DPP)

What is the DPP? An upcoming EU requirement under the ESPR. It is a "digital twin" (QR/NFC) containing verifiable data across the entire value chain (origins, carbon footprint, repairability).

Why should a CMO or CEO care?

1. Compliance

Without a DPP, you cannot legally sell in the EU market. It's the ultimate barrier to entry for non-compliant competitors.

2. Storytelling

Bulletproof evidence against Greenwashing. It proves sustainability claims and mathematically justifies a Premium Price.

3. Consumer Data

When a 3rd-hand owner scans the QR to learn how to repair it, the brand captures data on a "ghost" customer.

Why PaaS needs DPPs

The Data Dilemma

You cannot profitably manage a circular asset over its lifetime if you are blind to its data.

The Operational Risk

The Scenario:
You lease a product to 5 different users over a 10-year period.

The Blind Spots:
- ✗ What exact parts were replaced in year 3?
- ✗ What is its accumulated carbon footprint?
- ✗ Is it still safe to refurbish and lease again?

Result: Extreme financial and legal liability.

The DPP Backbone

What is it?
A mandatory EU "digital twin" that tracks the physical asset across its entire lifecycle.

The Capabilities:
- ✓ Immutable ledger of repairs and materials.
- ✓ Real-time proof of ESG compliance.

Result: The firm can safely scale an OPEX (PaaS) model with total traceability.

Anatomy of a Digital Product Passport

A DPP bridges the physical and digital worlds, creating a single source of truth across the entire value chain.

The Physical Asset

Scanned via Smartphone (NFC/QR) by the Consumer, Mechanic, or Recycler.

Unlocks Data Layers

1. Origin & Composition

Data: What is it made of? Where was the lithium mined? Exact carbon footprint.

✔ *Crucial for Green Claims validation and EU ESG audits.*

2. Usage & Maintenance Log

Data: Who owned it? When was the battery officially replaced?

✔ *Destroys Information Asymmetry and heavily boosts resale value.*

3. End-of-Life Routing

Data: Blueprint instructions for the recycling facility.

✔ *How to dismantle safely to extract high-value raw materials.*

Case Study: Michelin Fleet Solutions (B2B PaaS)

How the world's leading tire manufacturer used Data
to completely decouple revenue growth from resource extraction.

The Linear Trap (CAPEX)

The Flaw: Michelin's R&D made tires so durable that fleets bought fewer of them. R&D excellence was actively destroying their sales volume, while cheaper competitors stole market share.

The PaaS Pivot (OPEX)

The Shift: They stopped selling tires and started selling **"Kilometers Driven"**. Michelin retains ownership of the asset. The transport client pays a predictable monthly fee based on distance.

Data Enabler (Early DPP)

IoT Sensors:
Michelin embedded connected sensors inside the tires to track wear, pressure, and temperature in real-time.

Value Creation:
They use this data to optimize maintenance proactively and prevent blowouts, saving the client massive fuel and downtime costs.

Business ROI:
Michelin achieved pure **Decoupling**. By extending the life of the tire, they increase their own profit margin without needing to manufacture a new physical unit.

The Boardroom Debate: Project "Velocity"

The Scenario: Premium E-Bikes

You are consultants for **"Velocity"**, a premium E-Bike brand. The EU just passed strict Right-to-Repair laws. Selling bikes and ignoring their end-of-life is now illegal.

Group A

"The Use-Oriented Strategy" (Leasing)

The Pitch:
Transition to a monthly subscription model.

Your Core Tasks:
- ✔ How will you finance the heavy CAPEX?
- ✔ How will you handle reverse logistics?
- ✔ How will the DPP track the fleet's health?

Group B

"The Product-Oriented Strategy"

The Pitch:
Keep selling bikes, but build a high-margin repair and upgrade ecosystem.

Your Core Tasks:
- ✔ How to monetize the Right-to-Repair?
- ✔ How to incentivize part upgrades?
- ✔ How will the DPP build brand loyalty?

Executive Summary: The 3 Pillars of Week 5

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The 20% of knowledge that drives 80% of your results.

Final strategic takeaways for Servitization and Traceability.

1. The Transactional Flaw

The Transactional Model (Volume) is financially unsustainable due to severe margin erosion and new EU regulations.

2. Servitization (PaaS)

PaaS aligns corporate profit with planetary boundaries by decoupling revenue growth from pure physical resource extraction.

3. Digital Passports

DPPs are not just a legal requirement; they are the ultimate tool to eliminate Information Asymmetry and build Customer Lifetime Value (CLV).

Thank You.

Cristhian Rodríguez Revilla, PhD
Sustainable Development — Circular Economy
cristhian.rodriguezrevilla@ceu.es